Determination of the Absolute Zero Point

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Abstract

abstract

1 Introduction

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2 Experiment

experiment

3 Results

3.1 Calibration

For the calibration we measured the laboratory temperature t_l. We also measured the reading from the barometer $\tilde{p_l}$. Aswell as the output-signal of the pressure sensor when exposed to laboratory pressure U_l. Then we measured the output-signal at low pressure U_2. This gave:

$$\begin{aligned} t_l &= (22 \pm 1)°C \\ \tilde{p_l} &= (716 \pm 1)\ \text{mmHg} \\ U_l &= (128.0 \pm 0.2)\ \text{mV} \\ U_2 &= (-5.3 \pm 0.2)\ \text{mV} \end{aligned} \tag{3.1}$$

3.2 Absolute Zero

The measurements to determine the absolute zero temperature are the output signals of the pressure sensor. We use U_k for boiling water and U_e for freezing water. The results are:

$$\begin{aligned} U_k &= (127.6 \pm 0.2)\ \text{mV} \\ U_e &= (92.5 \pm 0.2)\ \text{mV} \end{aligned} \tag{3.2}$$

3.3 Liquid Nitrogen

The measurment for liquid Nitrogen (ln2) is U_{ln2}. This is the output signal from the pressure sensor when submerged in boiling liquid Nitrogen.

$$U_{\text{ln2}} = (24.7 \pm 0.2)\ \text{mV} \tag{3.3}$$

4 Data Analysis

4.1 Calibration

From the barometer reading $\tilde{p_l}$ and laboratory temperature t_l we calculate the pressure in the laboratory p_l

$$p_l = (\tilde{p_l} - \text{correction}(t_l)) * \frac{133.322}{100} = ((716 \pm 1) - 3) * \frac{133.322}{100} = (951 \pm 1)\ \text{mbar} \tag{4.1}$$

We used the value of U_k for U_l. so $U_l = (127.6 \pm 0.2)\ \text{mV}$ (discussion). The value for the low pressure is per assumption:

$$p_2 = (0.1 \pm 0.1)\ \text{mbar} \tag{4.2}$$

From here we fitted a linear function of the form:

$$p(U) = p_0 + CU \tag{4.3}$$

through (p_l, U_l) and (p_2, U_2). We calculated:

$$
\begin{aligned}
C &= \frac{p_l - p_2}{U_l - U_2} = (7.16 \pm 0.02)\frac{\text{bar}}{V} \\
p_0 &= p_l - \frac{p_l - p_2}{1 - \frac{U_2}{U_l}} = (0.038 \pm 0.001)\ \text{bar}
\end{aligned}
\tag{4.4}
$$

for the error propagation we always used a function of the starting parameters to calculate the values. We used the scipy Jacobian function to calculate all the derivatives. So for C it is:

$$
\begin{aligned}
z &= (p_l, p_2, U_l, U_2) \\
\sigma_z &= \left(\sigma_{p_l}, \sigma_{p_2}, \sigma_{U_l}, \sigma_{U_2}\right) = (0.001\ \text{bar}, 0.0001\ \text{bar}, 0.2\ \text{mV}, 0.2\ \text{mV}) \\
f_c(z) &:= \frac{p_l - p_2}{U_l - U_2} \\
\sigma_c &= \sqrt{\left(J_f(z)^2\right)_{1\text{x}4} (\sigma_z^2)_{4\text{x}1}} = 0.02\frac{\text{bar}}{V}
\end{aligned}
\tag{4.5}
$$

where $\sigma_z^2 = (\sigma_{z,1}^2, \sigma_{z,2}^2, \sigma_{z,3}^2, \sigma_{z,4}^2)$. And the same for the Jacobian $J_f(z)$. We used the same prcedure for all propagated errors. We always made the function depend only on the measured values and the values assumed and not on previously caluclated values. We did this to minimize the effect of correlation. Since previously calculated values depend on same starting parameters they correlate. To account for correlation the error propagation gets more complicated and one has to estimate the correlation between the values. This is why we always directly calculated from the original values.

The calibration gives the following calibration-curve:

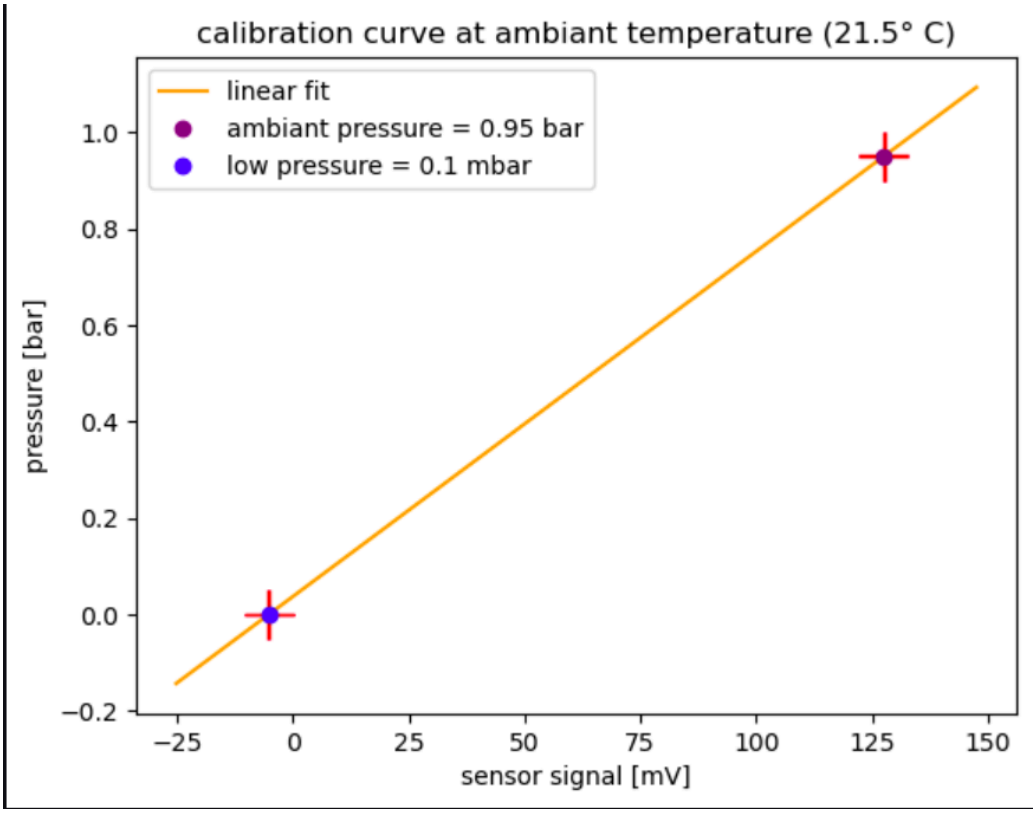


Figure 1: The two calibration points. (p_l, U_l) is purple and (p_2, U_2) is blue. Orange line is $p(U) = p_0 + CU$

4.2 Absolute Zero

From the calibration curve and U_k and U_e we obtain the pressure p_k and p_e at boiling temperature (water) t_k and at freezing temperature t_e.

$$
\begin{aligned}
p_k &= p(U_k) = (951 \pm 2)\ \text{mV} \\
p_e &= p(U_e) = (700 \pm 2)\ \text{mV} \\
t_k &= (98 \pm 1)°C \\
t_e &= (0 \pm 1)°C
\end{aligned}
\tag{4.6}
$$

We obtained t_k through a table in the laboratory where the boiling temperature of water is denoted in dependance of p_l. $t_e = 0$ per assumption. We can now also plot the pressure/temperature dependance:

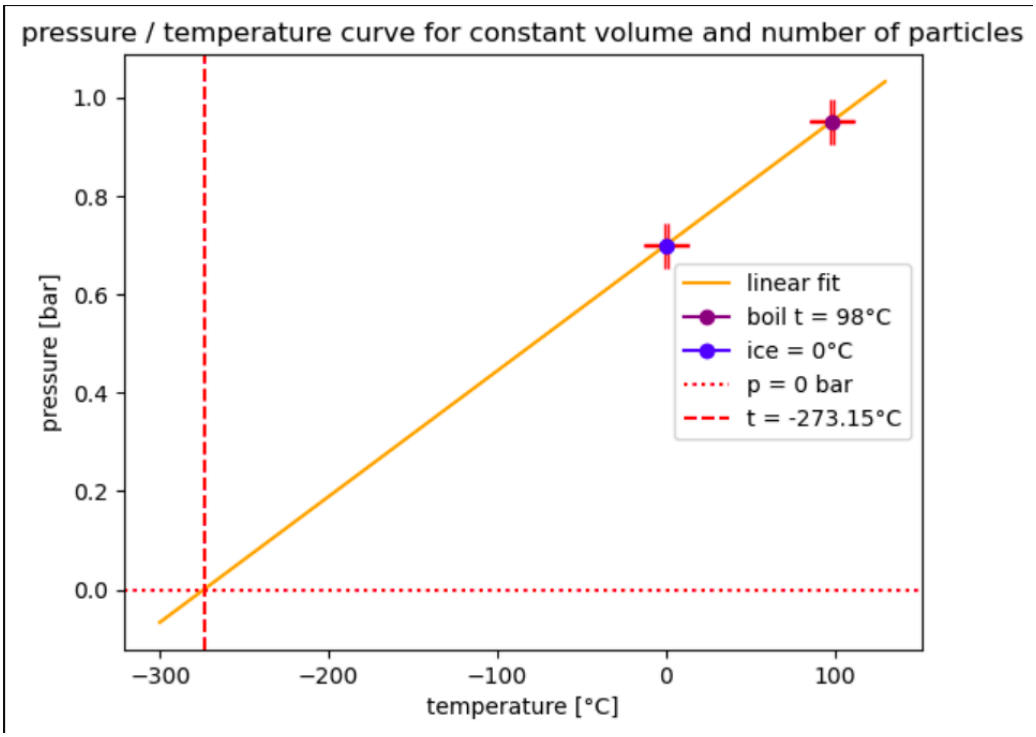


Figure 2:

From the obtained values we can caluclated the temperature of absolute zero. t_0' is a simpler approximative calculation. t_0 is more accurate since it takes some parazitic volume into account.

$$
t_0' = -\frac{p_e}{p_k - p_e} t_k = (-274 \pm 3)°C \qquad \text{manual} \tag{4.7}
$$

we define:

$$
\begin{aligned}
\varepsilon = 10^{-3}, \quad \gamma = 10^{-5}\ /°\text{C} \\
a = (1+\varepsilon)p_e - (1+\varepsilon+\gamma t_k)p_k \\
b = \varepsilon(p_k - p_e)t_k + (1+\gamma t_k)p_k t_l - p_e(t_l + t_k) \\
c = p_e t_l t_k \qquad \text{manual}
\end{aligned}
\tag{4.8}
$$

this gives:

$$
t_0 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} = (-273 \pm 5)°C \qquad \text{manual} \tag{4.9}
$$

4.3 Liquid Nitrogen

From the calibration curve and the measurment U_{ln2} we obtain the value for the pressure p_{ln2} at temperature of liquid nitrogen (t_{ln2})

$$
p_{\text{ln2}} = (214 \pm 1)\ \text{mBar} \tag{4.10}
$$

Again there is a simpler approximation for the temperature of liquid Nitrogen (t'_{ln2}). Aswell as a more complicated and more accurate value t_{ln2}. We have:

$$
t'_{\text{ln2}} = \left(1 - \frac{p_{\text{ln2}}}{p_e}\right) t_0 = (-189 \pm 4)°C \tag{4.11}
$$

we define:

$$\begin{aligned} \varepsilon &= 10^{-3}, \quad \gamma = 10^{-5} \ /^{\circ}\mathrm{C} \\ A &= \frac{p_e}{t_e - t_0} + \varepsilon \frac{p_e - p_{\mathrm{ln}2}}{t_l - t_0} \end{aligned} \tag{4.12}$$

this gives:

$$t_{\mathrm{ln}2} = \frac{A t_0 + p_{\mathrm{ln}2}}{A - \gamma p_{\mathrm{ln}2}} = (-189 \pm 4)^{\circ} C \tag{4.13}$$

5 Discussion

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6 Conclusion

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A Appendix

A.1 python skript

Bibliography